A UNIFIED DEEP-LEARNING FRAMEWORK FOR SCHWARZSCHILD QUASI-NORMAL-MODE EXTRACTION

Executive Summary

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Motivation & Scientific Justification

The ringdown of a perturbed black hole is a superposition of damped sinusoids, the quasinormal modes (QNMs), whose frequencies and damping times are fixed entirely by the black hole's mass and spin [1]. Measuring them from gravitational-wave data and comparing against the predictions of general relativity is among the cleanest strong-field tests available, and has been an observational programme since the ringdown analysis of GW150914 [2]. The precision of such tests rests on the accuracy of the underlying waveform and on the time interval used to fit its damped modal content [3,4].

Physics-informed neural networks (PINNs) solve differential equations by minimising the equation residual directly; the only prior PINN treatment of the time-domain Schwarzschild problem, however, reports a 28% relative waveform error that dominates its QNM error budget, with a fit window fixed by hand [5]. This project first tests whether that calculation can be reproduced and improved. It then addresses the per-instance training cost of a PINN with a hybrid Fourier neural operator (hFNO), which learns to correct a coarse numerical trajectory across a family of initial data. The same coarse-prior-plus-residual framework is subsequently adapted to the complex Teukolsky field of rotating Kerr black holes [9].

Methodology

Four numerical tests are evaluated. (i) A *reproduction* PINN, architecture-matched to the prior study [5], establishes a trustworthy baseline. (ii) An *enhanced forward* PINN doubles the quasi-Newton optimisation budget and adds residual-greedy sampling, which concentrates training points where the equation residual is largest. (iii) The Schwarzschild hFNO uses a $k = 4$ finite-difference trajectory as an additive prior and a 1.13-million-parameter FNO [6] to predict its discretisation error, following the solver-correction literature [7,8]. Its label-free target is formed by Richardson extrapolation of the $k = 4$ and $k = 2$ solves, and a prior-gradient gate confines the correction to under-resolved wavefronts. Coarsening both space and time by four reduces the finite-difference point-update count by approximately a factor of sixteen; the total wall-clock speedup has not been measured. (iv) For Kerr, the framework is adapted to a complex hyperboloidal Teukolsky field and retrained

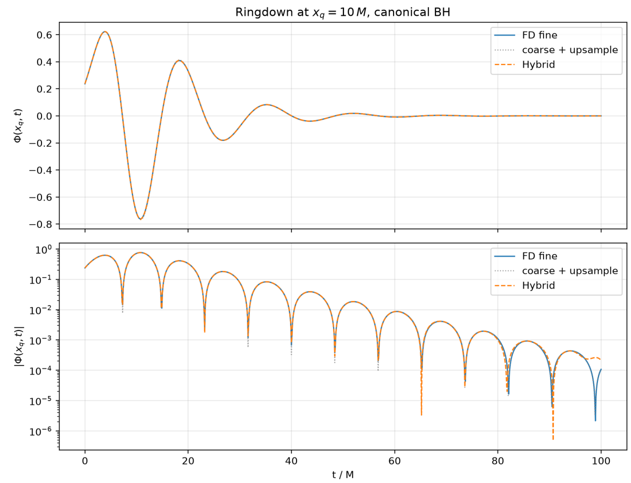

Figure 1: Ringdown waveform at the observer for the benchmark configuration, on linear (top) and logarithmic (bottom) scales: the Schwarzschild hFNO (dashed) overlaid on the fine finite-difference reference (solid) and the coarse prior (dotted).

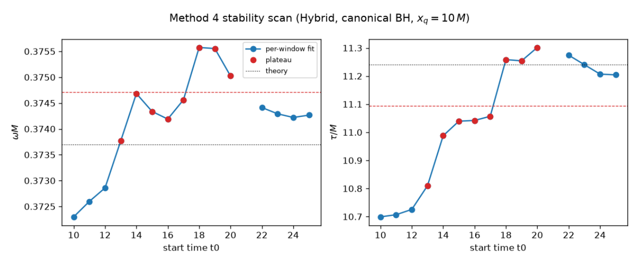

Figure 2: The plateau scan underlying the extraction protocol: recovered frequency (left) and damping time (right) as the fit-window start time is swept for the Schwarzschild hFNO. Red markers form the selected contiguous block; dotted lines mark the continued-fraction values.

across spin. The deployment prior uses $N = 101$, while independent $N = 201$ and $N = 401$ fields form the Richardson target and an $N = 801$ field is retained for evaluation.

Each field is analysed with the post-hoc extraction framework of the report. Five real-signal estimators are used for Schwarzschild, with complex-signal extensions for Kerr. One- and two-edge plateau scans sweep the fit window and report the variation of the fitted frequency and damping time across its most stable contiguous block (Figure 2).

Key Findings

The reproduction reduces the prior study's waveform error by an order of magnitude on both of its benchmark potentials (relative L^2 error 2.61% versus 28.06%, and 2.67% versus 23.59%). The enhanced forward PINN reaches 0.58%, from which the best-of-suite estimates recover the fundamental frequency to 0.029% and damping time to 1.99% relative to Leaver's values.

Across the 100-configuration Schwarzschild test population, the hFNO reduces the mean field error from 13.39% to 1.85% and the field MSE by a factor of 46, without using fine-grid fields in its training loss. Its population frequency error is 0.152%. After adaptation and separate retraining, the Kerr hFNO reduces the median field error from 59.5% to 0.78% across 128 test configurations. In the high-spin bin, the median frequency error falls from 4.71% for the coarse prior to 0.38%. The damping time is the exception: its population error increases from 0.52% to 1.52%, consistent with a residual in the low-amplitude late ringdown.

Recommendation & Next Steps

The immediate priorities are an objective that gives greater weight to the late-time envelope, and an end-to-end runtime comparison that includes interpolation and neural inference. Tests outside the training ranges and under changes of domain and resolution are also needed before broader deployment. A later inference study could combine the hFNO waveform family with reduced-order likelihood methods to test whether rapid field generation produces a practical gain in Bayesian parameter estimation.

Research Impact & Conclusion

The central result is that learning a correction to a coarse numerical prior gives accurate fields across the tested Schwarzschild and Kerr parameter families without fine-grid fields in the training loss. The classical solver retains propagation and boundary structure, while the operator concentrates on the discretisation defect. The Kerr damping-time result also identifies the present limit clearly: global field accuracy does not by itself guarantee an accurate low-amplitude late-time envelope.

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